

Deliverables (to be submitted on Quercus):

All Python source code / answers in a Jupyter Notebook (*.ipynb).

Any additional documents (Word, PDF, CSV, TXT, HTML, etc.) as required.

AI Appendix at the end of the notebook (only if you used generative AI; see “Use of Generative AI” below).

Rotman Investment Club

Project Overview

In this group assignment, you will be working with the Rotman Investment Club to leverage recent advancements in artificial intelligence. The goal is to develop an automated pipeline that can clean and parse raw SEC 10-K and 10-Q forms.

Background

Financial disclosures, such as annual reports (SEC Form 10-K) and quarterly reports (SEC Form 10-Q), provide comprehensive self-evaluations of a company's recent performance. These reports are essential for valuing stocks and bonds. While financial ratios from these reports are readily available, they only scratch the surface of the information contained within these documents. The narratives provided in these disclosures contain crucial information that can't be gleaned from financial ratios alone.

Given the sheer volume of digitized financial reports, it's impossible for humans to process all the information. For instance, on August 8th, 2023, 511 companies submitted 10-K and 10-Q forms to the SEC EDGAR database, followed by 589 new submissions on August 9th, 2023. It's unrealistic for investment analysts to read and digest over 1,000 reports in two days. However, machines can scan these reports in a timely manner and retrieve predefined signals.

Every day, the Rotman Investment Club retrieves the latest SEC Forms 10-K and 10-Q from the SEC EDGAR database. These forms are submitted as txt and XBRL files. While XBRL files can easily be rendered into a webpage, they are challenging for humans and machines to read due to different text formatting, data labels, styles, and conventions. These files can also be downloaded in plain HTML format, but they contain numerous HTML tags for formatting.

Project Objectives

The aim of this project is to develop a pipeline that can clean and parse downloaded files of SEC Form 10-K and 10-Q. Specifically, the tasks include:

- Removing tags and retaining only text and tables.
- Extracting individual sections from 10-K and 10-Q, including but not limited to Business, Risk Factors, Management's Discussion and Analysis, etc.

Data

You will be working with historical SEC Form 10-K and Form 10-Q data. All 10-K and 10-Q data are downloaded. The size of all historical data, dating back to the 1990s, is around 2TB.

Benefits of the Project

Upon completion of this project, the Rotman Investment Club will have a machine-readable 10-K and 10-Q dataset, broken down to section levels. As a participant in this project, you will:

- Gain detailed knowledge about the SEC EDGAR database.
- Learn and develop cutting-edge text mining techniques to parse digitized financial reports and extract specific information from filings.
- Gain experience working with big and unstructured text data in the finance and investment field.

Proof of Concept

To develop a proof-of-concept prototype, only use the most recent annual reports (i.e., SEC Form 10-K) for the following ten US companies:

1. Apple Inc.: A multinational technology company headquartered in Cupertino, California, United States.
2. Microsoft Corporation: A US American multinational technology company headquartered in Redmond, Washington, United States.
3. Amazon.com, Inc.: A US American multinational technology company based in Seattle, Washington, United States.
4. Tesla, Inc.: A US American electric vehicle and clean energy company headquartered in Austin, Texas, United States.
5. Meta Platforms, Inc. (formerly Facebook, Inc.): A US American social media conglomerate based in Menlo Park, California, United States.
6. Alphabet Inc.: A US American multinational conglomerate headquartered in Mountain View, California, United States.
7. Johnson & Johnson: A US American multinational corporation headquartered in New Brunswick, New Jersey, United States.
8. The Coca-Cola Company: A US American multinational beverage corporation headquartered in Atlanta, Georgia, United States.
9. Procter & Gamble Co.: A US American multinational consumer goods corporation headquartered in Cincinnati, Ohio, United States.
10. Walmart Inc.: A US American multinational retail corporation headquartered in Bentonville, Arkansas, United States.

To download them from the web, go to the EDGAR website, in particular to their company search page <https://www.sec.gov/edgar/searchedgar/companysearch>, and once you have found their respective *latest* 10-K Form and open it in plain HTML. Save those plain HTML files to your project directory. Those HTML files should be the only HTML files in your project directory!

Submit those files as part of the assignment.

All the remaining steps have to be performed using Python!

The code should be submitted in form of a single Jupyter Notebook (*.ipynb).

Hint: You should be using a mix of string methods and/or the power of RegExes as needed to provide short and concise code. You should not do a “manual” case by case implementation nor use BeautifulSoup (please see Bonus).

When loading the individual reports, you should use a loop that iterates over the HTML files in your project folder and you should not “hard”-code the names into a list or even load them one by one by manually changing the code.

Hint:

```
from pathlib import Path
# Use the current directory
current_directory = Path.cwd()
# Iterate through all HTML files in the current directory
for html_file in current_directory.glob("*.html"):
    with html_file.open('r', encoding='utf-8') as file:
        content = file.read()
        print(f"Loaded {html_file.name}:")
```

For each HTML file that you are processing, the HTML tags need to be removed and only cleaned text should be kept. Do this using string methods and/or regular expressions.

Run the cleaned text through Norvig’s spell checker.

Instead of using the over 130-year-old (published on 14 October 1892) Sherlock Holmes stories (which are fun to read, by the way! 😊), use a dictionary instead as the underlying corpus. For example, you could use Webster’s Unabridged Dictionary as given on the Gutenberg Project website <https://www.gutenberg.org/cache/epub/29765/pg29765.txt>. What are the challenges of using this spell checker? How do you address them?

For each company, save the clean-text 10-K Form into a separate plain text (*.txt) file. Submit those files as part of the assignment as well.

For the prototype as a proof of concept collect the following content, which should be filtered out:

1. The date of the fiscal year end (ensure it is formatted in ISO-format).
2. The content of “Item 3. LEGAL PROCEEDINGS”.
3. The date of signature(s) (ensure it is formatted in ISO-format).
4. Who signed the report? If there are multiple signatures, all of them have to be listed (comma separated). Do not include the audit firm – in case it is given in the report.

Create a simple Pandas DataFrame with several columns: the company name and the content collected above.

With `df.to_csv("legalproceedings.csv", index=False)`, you should write the DataFrame into a plain text CSV file, which you should also submit.

Hint: For efficiency reasons, you might want to write the data into a Python dictionary first, and then transform this into a Pandas DataFrame. This is faster than appending row by row of new data to a Pandas DataFrame.

Verification (required)

Automated extraction can look right and still be wrong. For each of the ten companies, show that your results are correct:

(a) Print the first and last 200 characters of your extracted “Item 3. LEGAL PROCEEDINGS” text and confirm that the section starts and ends where it should (the next item heading must not be included).

(b) Build a check table that compares your extracted fiscal-year-end date, signature date(s) and signers with the values you read in the filing yourself, and mark every mismatch.

(c) Explain every mismatch, and either fix your code or explain why it cannot be fixed with your approach.

This verification is graded, regardless of how your code was written.

Bonus (1 bonus point): Use the Python package Beautiful Soup to extract the data as described above, and compare its results with those of your string/regex version for all ten companies. Where do the two versions disagree, and which one is right?

Team, Submission, and Academic Integrity:

This assignment is completed in teams of 3–4 (preferably 4). One team member submits all files for the team on Quercus by Friday, October 23, 2026, 23:59.

Your Jupyter Notebook must run from top to bottom on a fresh kernel (Kernel → Restart & Run All) without errors, reading all input files from your project folder via relative paths, so that every result you report can be reproduced.

Late submissions will normally be penalized by 20% if the assignment is not received on the specified date, at the specified time. A further penalty of 10% will be applied for each subsequent day.

Source code submitted for the group assignments may be reviewed using MOSS (Measure of Software Similarity), an automated system for detecting similarity among programs.

Use of Generative AI:

Generative-AI tools (ChatGPT, Claude, Copilot, and similar assistants) are permitted on this assignment, subject to the policy in the course outline. This course does not teach how to use these tools; whether and how you use them is your team’s decision and responsibility.

Disclosure. In a brief AI Appendix at the end of your notebook, state which tool(s) you used and for which parts of the work — for example, “Used ChatGPT to draft an initial version of our text-cleaning function; verified, tested, and edited by hand.” Undisclosed use of generative AI on graded work may be treated as an unauthorized aid.

Responsibility. You are accountable for everything you submit, including any AI-generated content. These tools readily produce plausible-looking output that is wrong, so you are expected to verify, test, and understand any code or analysis you turn in. “The AI gave me that answer” is not an acceptable defence.

Be exam-ready. The final examination is written in person without any generative-AI tools. Use this assignment to practise the methods yourselves.